- List = [ ] 可变
  Tuple = ( ) 不可变
  Set = { } 可变
  Dictionary = { } 不可变

PANDAS

- DataFrame

Method	Use Case	Works On
.apply(func)	Apply to a column or across rows	Series or DataFrame
.apply(func, axis=1)	Use row vales to calculate new result	DataFrame
.map(func)	Apply element-wise to all values	DataFrame

- break = 马上 exit.
- continue = move on the next iteration
  pass = do nothing in the block.

- List methods (Tuple methods)

`append(x)` adds x to the end of the list
`count(x)` counts how many times x appears in the list
`extend(L)` adds the elements in list L to the end of the original list
`index(x)` returns the index of the first element of the list to match x
`insert(i, x)` inserts element x at location i in the list, moving everything else along
`pop(i)` removes the item at index i
`remove(x)` deletes the first element that matches x
`reverse()` reverses the order of the list
`sort()` we've already seen

- Set methods

Function	Alternate Syntax	Description
a.add(x)	N/A	Add element x to the set a
a.clear()	N/A	Reset the set a to an empty state, discarding all of its elements
a.remove(x)	N/A	Remove element x from the set a
a.pop()	N/A	Remove an arbitrary element from the set a, raising KeyError if the set is empty
a.union(b)	a | b	All of the unique elements in a and b
a.update(b)	a |= b	Set the contents of a to be the union of the elements in a and b
a.intersection(b)	a & b	All of the elements in *both* a and b
a.intersection_update(b)	a &= b	Set the contents of a to be the intersection of the elements in a and b

- Dict methods
  - key() return key
  - value() return value
  - items() return both
- .map(function, iterable)
  把某個函數套用到 iterable
- lambda x = expression.
- Numpy Array Attribute

Function	Description
ndarray.shape	Tuple of array dimensions.
ndarray.ndim	Number of array dimensions.
ndarray.itemsize	Length of one array element in bytes.
ndarray.size	Number of elements in the array.
ndarray.dtype	Data-type of the array's elements.
ndarray.T	The transposed array.
ndarray.real	The real part of the array.
ndarray.imag	The imaginary part of the array.
ndarray.flags	Information about the memory layout of the array.

- Unary Function

Function	Description
abs, fabs	Compute the absolute value element-wise for integer, floating-point, or complex values
sqrt	Compute the square root of each element (equivalent to arr ** 0.5)
square	Compute the square of each element (equivalent to arr ** 2)
exp	Compute the exponent e^x of each element
log, log10, log2, log1p	Natural logarithm (base *e*), log base 10, log base 2, and log(1 + x), respectively
sign	Compute the sign of each element: 1 (positive), 0 (zero), or –1 (negative)
ceil	Compute the ceiling of each element (i.e., the smallest integer greater than or equal to that number)
floor	Compute the floor of each element (i.e., the largest integer less than or equal to each element)

- Binary Function

Function	Description
add	Add corresponding elements in arrays
subtract	Subtract elements in second array from first array
multiply	Multiply array elements
divide, floor_divide	Divide or floor divide (truncating the remainder)
power	Raise elements in first array to powers indicated in second array
maximum, fmax	Element-wise maximum; fmax ignores NaN
minimum, fmin	Element-wise minimum; fmin ignores NaN

- loc[ ] 用 label name 取
  (用 " " 存取, 可Boolean)
  iloc[ ] 用 data 位置取
- access to column
  _____ ["column name"]
- filter = Boolean
  用做像 df[df[" "] ___]
- Manipulation
  drop { axis=1 ↓ drop 一直行.
  axis=0 → drop 一橫列.
- rename (columns = {" ": " "})
- sort_values (___, ascending=T or F)
  ↳ 卡.sort() 卡 sorted()
- apply ((lambda x: _____)
  apply (def)
- map (_____)

- 常態分佈 = heights, test scores, blood pressure
- Right skewed = Income
  sales transaction, Hospital stay.
- Left skewed = Retire age.
  test scores (easy), death age, employer tenure
- Bimodal = Income level, marriage age, study hour. ↑
- log-normal = stock price, population sizes. company size
- Truncated = Age data, Income data. Testscore
- Poisson = defects, number of calls, websites hits per minute
- exp = Time between arrivals.
  Light bulbs failure.

Outliers
- z-score: 4-5 SD.
- IQR = 25+1.5IQR,...
- QQ: assess how well follow the theory / identify deviate / assumption of normality.

- interaction / confounding
  ① joint relationship.
  → need to lower dimension
  Remove / combine Variable
  ② hidden relationship
  → need to include
- multicollinearity
  - 回歸係數不穩定
  - 預測的表現↓
  → Remove / combine related
  → dimension reduction
- High leverage observation
  - 影響 Regression a lot

productivity.

- activites performed (uni)
  - Identifying and characterizing missing data
  - Detection outliers
  - Assessing the distributions of data
- main objective (bi)
  - Identifying, characterizing and evaluating correlations between variables.
  - Analyzing how different categories of one variable behave in relation to another variable (e.g., how different treatments affect recovery times in clinical trials)
- main objectives (bi diagnos)
  - Assess assumptions (for example, linearity, independency of observations, etc.)
  - Detect and manage bivariate and multivariate outliers
  - Detect and manage multicollinearity
- variance (large/small?
  - Comparison: compare it to the variance of other similar datasets
  - Visualization: visualize the distribution of your data using histograms and compare it with theoretical distribution such as normal curve
  - Coefficient of Variance
- 3 reason for EDA
  - To understand as thoroughly as possible the dataset and what each variable represents
  - To identify data quality issues and other issues that will inform the data preparation activities
  - To enhance communications between the business stakeholders of the analytics project and the technical team

M3-1 Ingest/Integrate

- JSON [{object} key = value pair, key ....
  [array] 重複性的data.

① Prep Step 1

① Inspect Structure
- .shape - .columns / .dtypes - .info()

② Preview Sample rows
- .head() - .sample(n) = random rows
- .tail() = 同步等待取排取的info.

③ Check for missing values
- .isnull().sum() = Total missing value
- .notnull().mean() = proportion of non-missing value

④ Assess duplicates
- .duplicated.sum()
- df[df.duplicated(subset = key columns)]
  = check duplicates on key

※ print("字{查入的df}") ↳ 大括号.

⑤ Spot common formatting or type issues

② Prep Step 2

- to_numeric(____, errors= ____, downcast = ____)
  - raise = 遇到錯值就報錯.
  - coerce = 錯值 = NaN.
  - ignore = 有錯值就全不轉換
  - ↳ 縮小資料型式
- pd.to_datetime
- reorder column ⇒ q2 = q2[q1.columns]
- drop dummy header ⇒ q2 = q2[q2["column"] != "Content"]
- .rename(columns = {"原名": "新名"}, inplace = ____)
  - True 直接修改原物件
  - False 回傳新的副本
- .str.strip() / .str.upper()
- .drop_duplicates

③ Prep Step 3

Vertical = 直向的加入 same column 更多 data
= pd.concat( ), axis = 0
- axis = 0 向上下接 / add more row
- axis = 1 向左右接 / add more column

Horizontal = 橫向加入更多屬性的 column
= pd.merge(被加入者, 加入者, ...)
- on: left_on = ____, right_on = ____
  (不同table名稱不同時)
- how = inner ... (連結方法)

M3-2 Reshape

① Tall → wide = pivot + (reset_index())

df.pivot(index = 'column_name(s)', columns = 'column_name', values = 'column_name(s)')
- index: Column(s) to use as the new row index
- columns: Column whose unique values become the new column header
- values: Column(s) whose values fill the new table cells. If omitted, all remaining columns are used

② wide → Tall = melt / groupby/split/reorder

df.melt(id_vars = 'column_names', value_vars = 'column_names', var_name = 'new column name', value_name = 'new column name', col_level = 'column_name', ignore_index=True/False)

Parameter	Description
id_vars 保留	Column that will become the "identifier" column for the new DataFrame
value_vars → 變長	Columns to convert from wide to tall. If none, all columns will be converted
var_name 新 column名	Name of new column that will store the former column names
value_name 值	Name of new column that will store the unpivoted values
col_level	For DataFrames with MultiIndex columns – specifies the level to melt
ignore_index	If True (default), the result index is reset to 0…n-1

Tidy vs. Spread

Tidy Data (Tall Format)	Spread Data (Wide Format)
Each variable has its own column	Multiple variables embedded across multiple columns
Each observation has its own row	One row may contain multiple observations for the same entity
Each type of observational unit has its own table	Different observation types may be mixed in the same table

python reshaping function

When you need to...	Method	Example
Wide → Tall (tidy)	**.melt()**	df.melt(id_vars=['Store'], var_name='Month', value_name='Sales')
Tall → Wide (single-level)	**.pivot()**	df.pivot(index='Store', columns='Month', values='Sales')
Tall → Wide + aggregation	**.pivot_table()**	df.pivot_table(index='Store', columns='Month', values='Sales', aggfunc='sum')
Swap rows ↔ columns	**.transpose() / .T**	df.T
MultiIndex rows ↔ columns	**.stack() / .unstack()**	df.unstack(level='Month')
Make columns the index	**.set_index()**	df.set_index(['Region', 'Store'])
Move index to columns	**.reset_index()**	df.reset_index()
Sort by index or values	**.sort_index() / .sort_values()**	df.sort_index(axis=1)
Remove MultiIndex level	**.droplevel()**	df.columns = df.columns.droplevel(0)
Rename columns/index	**.rename() / .rename_axis()**	df.rename(columns={'old':'new'})
Clean column names	**.str.replace() / .str.strip()**	df.columns = df.columns.str.replace(' ','_')

When you need to...	Method	Example
Clean column names	**.str.replace() / .str.strip()**	df.columns = df.columns.str.replace(' ','_')
Flatten MultiIndex cols	**list comprehension**	df.columns = [f"{a}_{b}" for a,b in df.columns]
Split col into multiple	**.str.split()**	df[['Month','Year']] = df['Period'].str.split('-', expand=True)
Combine cols into one	**.str.cat()**	df['Full'] = df['First'].str.cat(df['Last'], sep=' ')
Create datetime from parts	**pd.to_datetime()**	pd.to_datetime(df[['Year','Month','Day']])
Aggregate pre-pivot	**.groupby() + .agg()**	df.groupby('Region')['Sales'].sum().reset_index()
Fill gaps after reshape	**.fillna()**	df.fillna(0)
Drop empty rows/cols	**.dropna()**	df.dropna(how='all')
Ensure all combos exist	**.reindex()**	df.reindex(new_index, fill_value=0)
Special: Structured reshape	**pd.wide_to_long()**	pd.wide_to_long(df, stubnames='Sales', i='id', j='Year')
Special: Split lists into rows	**.explode()**	df.explode('Tags')

EDA

identify pattern + Assess quality + Communication

- classification = for analysis / visualize / modeling.
  - Quan vs. Qual = 平均有無意義.
  - Ratio v.s. Interval = True zero V X
- Univariate
  - Descriptive = Location / variable
    skewness (-1 ~ 1) / kurtosis
  - Objectives =
    ① understand distribution
    ② Visually identify Central / spread / shape
    ③ detect outliers, patterns.
  - visualization
    histogram / Boxplot / Violin Plot
  - categorization
    Barchat / Pie chart.

Step	MAR	MNAR
Correlation with Observed Data	Missingness correlates with observed variables.	No correlation, or insufficient to explain missingness.
Logistic Regression	Missingness is predictable by observed variables.	Missingness is not predictable by observed variables.
Domain Knowledge	Missingness aligns with plausible external factors (e.g., age or income category).	Missingness tied to unobserved (missing) values (e.g., high income, severe symptoms).
Informative Missingness	Missingness does not convey additional information about the data.	Missingness itself is meaningful and provides insight into the unobserved data

MCAR	Low	Simple deletion or imputation; safe to ignore missingness
MAR	Medium	Use observed data for imputation or model adjustments
MNAR	High	Requires domain knowledge